

Only the Outermost Bound System Carries the Phantom: Hierarchical Ownership in a Λ -Tied MOND Law, Its Scorecard, and an Exactly Newtonian Wide-Binary Prediction for Gaia DR4

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Abstract

The programme’s central relation ties the MOND acceleration scale to the vacuum, $a_0 = \kappa c \sqrt{G \rho_\Lambda}$, with $\kappa = \frac{1}{2}$ fitted, not derived [Z26a]. This note reports a principle that decides *which systems the law acts on*, the scorecard it earns, and the prediction it makes for Gaia DR4 before the data exist. *Hierarchical ownership*: the law acts on a system’s own baryons only if that system is the outermost bound system. An accreted satellite keeps the dark component it owned at infall; a system that formed embedded in a host (a tidal dwarf, a globular cluster, collision debris, a wide binary, the Solar System) is Newtonian. The rule adds no constant and removes one, the Solar-System screening length. On the record’s population gates it passes 9 of 14, against 6 for the same law with the host’s external field. With the infall baryons its own logic calls for, the M31 satellites pass (1.2–1.6 σ), and the combined target law passes 42 of 48 harness gates. It fails two: the Milky Way’s ultra-faint satellites (3.5–3.8 σ once refereed, first quoted at 7–8 σ on statistical errors alone; the external-field reading fails them at 13 σ) and the external-field signal of Chae et al. (4.1–4.3 σ in their fits; 1.7–2.2 σ refit under the framework’s own law at the canonical scale). A local virial theorem explains the phantom’s shape: a cold component in local virial equilibrium around a point mass reproduces the kernel $\nu = \sqrt{1 + 1/y}$ exactly, with a_0 as its charge per unit baryonic mass; SPARC disfavours that kernel’s transition shape at about 2 σ . The dark sector’s edge meets a 5.4 σ tension between KiDS-1000 lensing and the cosmic cold budget. The same lensing machinery misfits standard Λ CDM halos by at least $\Delta\chi^2 = 50$, so the tension is not decidable there. The Local Group’s zero-velocity radius conflicts with standard halos at least as strongly as with the framework. *The prediction*: wide binaries are exactly Newtonian, $\gamma_v = 1.000$ on the pre-registered Gaia DR4 estimator, falsified at $\hat{\gamma} \geq 1.084$. The same law with the external field predicts 1.16–1.23 and is falsified by a Newtonian result. Gaia DR4 is scheduled for 2 December 2026.

1 The relation, the footings and the kernels

The relation is $a_0 = \kappa c \sqrt{G \rho_\Lambda}$ with $\kappa = \frac{1}{2}$, equivalently $\Lambda \ell_0^2 = 32\pi$ with $\ell_0 = c^2/a_0$. Its form is forced; its coefficient is fitted, and no action in the class the programme has built derives it [Z26a]. Every π in 32π is Einstein’s 8 π , so the relation’s one convention-free number is $1/\kappa^2 = 4$: the vacuum energy density is $4a_0^2/G$. No natural construction on the vacuum gives $\kappa = \frac{1}{2}$ (lane k05). The relation predicts a_0 flat in redshift [Z26c]. Wherever a_0 enters, every lane runs on two footings: canonical, $a_0 \approx 9.36 \times 10^{-11} \text{ m s}^{-2}$ from ρ_Λ ; and alternative, $\approx 1.13 \times 10^{-10} \text{ m s}^{-2}$ with the total density in place of ρ_Λ . Pairs written x/y are canonical/alternative.

The galaxy law is $g = \nu(g_N/a_0) g_N$. Two kernels are carried: the programme’s contract kernel ν_{mono} , which differs from the RAR function $\nu_{\text{RAR}}(y) = 1/(1 - e^{-\sqrt{y}})$ [MLS16] by at most 0.0104 dex; and $P2$, $\nu = \sqrt{1 + 1/y}$. With a_0 held at the framework’s value and one global stellar mass-to-light ratio profiled, the SPARC rotation curves [LMS16] are fitted to 0.100/0.099 dex rms by ν_{mono} and 0.108/0.104 dex by $P2$ (lane CFG4).

2 Hierarchical ownership

The framework’s target law (CFG4) treats the dark component as a cold fluid, with the cosmic amount $\Omega_c h^2 = 0.12$ fitted as in Λ CDM. Inside a bound region the component *is* the law’s phantom density; outside, it behaves as CDM. Counting a bound subsystem’s own phantom on top of its host’s would count the same cold matter twice. That forces a rule for which system owns the phantom (lane FG001):

- **Top-level systems** (an isolated galaxy, a group, the Local Group as a whole) obey the law, with no external-field effect.
- **Accreted systems** (satellites, cluster members, ultra-diffuse galaxies) keep the cold component they owned at infall. Their internal dynamics is the *isolated* law applied to their *infall* baryons.
- **Systems formed embedded** (tidal dwarfs, globular clusters, collision debris such as NGC 1052-DF2 and DF4, wide binaries, the Solar System) never owned a cold component and are Newtonian.

The rule adds no constant. It removes one: the screening length the programme’s other readings need for the Solar System. The Sun owns no phantom, and the Milky Way’s phantom tide at the Sun, $1.6\text{--}2.6 \times 10^{-31} \text{ s}^{-2}$, is $2\text{--}3 \times 10^4$

below Cassini’s quadrupole bound. It differs from standard MOND in a testable way. Standard MOND applies the law to every subsystem, with the host’s field entering through the external-field effect. Ownership applies it only to the outermost system, and a subsystem sees its host’s field only as a tide.

3 The scorecard

Table 1 scores the rule against the record’s population gates, next to the same law with the host’s external field (the programme’s previous reading). Each entry is a committed lane’s number; both footings are given where they differ.

gate	hierarchical ownership	law + external field
Cassini, no screening constant	pass (margin $2-3 \times 10^4$)	needs a screening length
Milky Way classical dSphs	$1.1/0.7\sigma$; with infall gas $0.4/0.1\sigma$	$3.5/3.3\sigma$
Local Volume dwarfs, host statistic	1.7σ	3.9σ
cluster-infall BTFR slope / zero point	0.1σ / 1.3σ	2.4σ / 0.8σ
NGC 1052-DF2 / DF4	0.0σ / 0.8σ	$2.9/3.1\sigma$ / 1.5σ
tidal dwarfs	pass	pass on χ^2 , worse by $\Delta\chi^2 \geq 5$
globular clusters	$M/L_V = 1.70$ (Newtonian)	needs $M/L_V \approx 1.0$
M31 dwarfs, LVD [Pac24]	$2.6/2.2\sigma$; with infall gas $1.6/1.2\sigma$	$6.0/5.7\sigma$
M31 dwarfs, Collins+13 [Col13]	$2.5/2.3\sigma$; with infall gas $1.4/1.2\sigma$	$5.8/5.6\sigma$
Milky Way ultra-faints	$8.0/7.5\sigma$; with infall gas 7.4σ ; refereed $3.8/3.5\sigma$	$13.2/12.8\sigma$
Chae et al. D1 / D2 [Cha20]	$4.1/4.3\sigma$ (the cost)	$0.8/0.0\sigma$

Table 1: The population gates (lanes FG001, CFG18). On the 14 gates as first scored, ownership passes 9 and the external-field reading 6, on both footings.

Infall baryons. FG001 first scored satellites with their present-day stars. Ram pressure has since removed their gas, and the rule’s own class statement asks for the baryons they carried at infall. Lane CFG18 draws each satellite’s infall gas-to-star ratio from its five nearest neighbours in stellar mass among 42 Local Volume field dwarfs with HI measurements, counting non-detections as zero gas (the conservative bracket), over 2000 realisations. The M31 samples then pass on both footings. Two pre-declared sub-hypotheses of FG001 failed and are kept as they fell: globular clusters do not discriminate, and the M31 LVD dwarfs sat just past the bar before the infall correction. The combined target law, CFG4 plus ownership, passes 42 of 48 gates of the programme’s harness (CFG19). The six failures are the ultra-faints and Chae’s D1/D2, each on both footings.

The two failures. The Milky Way’s ultra-faint satellites fall below the calibrated mass range and were quenched before they could hold gas, so the infall correction cannot reach them. The first scoring put them $7-8\sigma$ high, on statistical errors alone and without the nine systems that have only upper limits. The adversarial referee of lane CFG28 restores those limits with a Kaplan–Meier median and adds a coherent systematic floor (the stellar mass-to-light ratio and the estimator). The offset stays, $+0.32/+0.30$ dex, but at $3.8/3.5\sigma$. Tides do not explain it: systems beyond 80 kpc show the same offset. Measurement noise does not either: the best-measured systems show more. Lane CFG29 bounds unresolved binaries with published simulations; the offset persists where binaries are weakest, and two systems exceed any single-epoch binary ceiling. It is a real failure, smaller than first quoted. The external-field reading fails them harder.

Chae et al. detect an external-field effect in 153 SPARC galaxies [Cha20], and ownership predicts none for isolated top-level galaxies. Refitting all 153 with their exact priors and likelihood (lane CFG8, their numbers reproduced), but under the framework’s own law and scale, the median fitted field falls from their 4σ to 1.7σ ($P2$) and 2.2σ (ν_{mono}) on the canonical footing, and to $2.7-3.0\sigma$ on the alternative. Their low-acceleration downturn disappears under $P2$. The cost is reduced, not removed.

4 Why the phantom has its shape: a local virial theorem

Place a cold component around a point baryonic mass M , in isotropic Jeans equilibrium in the total field, and require every shell to be virialised in its own circular speed, $\sigma^2 = V_c^2/2$. The Jeans equation then gives $\rho = K/(r^3 g)$.

With $u = r^2 g$, the mass equation integrates to

$$u^2 = (GM)^2 + 4\pi G K r^2 \implies g = \sqrt{g_N^2 + a_0 g_N}, \quad a_0 = \frac{4\pi K}{M}, \quad (1)$$

which is exactly the kernel $P2$ at every radius (lane CFG9; verified to 9×10^{-8} over $g_N/a_0 \in [10^{-4}, 10^4]$, both footings). The solution of the ODE is unique, so $P2$ is the only law whose point-mass phantom is locally virialised at every radius. Among $\nu_\beta = (1 + y^{-\beta})^{1/2\beta}$, only $\beta = 1$ qualifies. Two equivalent statements follow. The component's pressure obeys $4\pi r^2 P = a_0 M/2$: its pressure force on any sphere is half of a_0 times the enclosed baryonic mass. And $P = a_0 g_N/(8\pi G)$ at every radius. a_0 is the component's charge per unit baryonic mass, which the framework ties to the vacuum.

The theorem fixes the shape only where the baryons look like a point mass. SPARC's transition region lies largely inside the baryons, and there the data prefer a softer transition. Calibrated against 60 noise realisations of each truth, $P2$ applied everywhere is disfavoured at $p \approx 0.03$ (about 2σ) on the canonical footing and at $p \approx 0.03$ – 0.13 on the alternative, while ν_{mono} is fully consistent (lanes CFG13, CFG14). The points in the point-mass regime do not discriminate. The theorem explains the kernel's asymptotics and its form around compact masses; it does not select the contract kernel.

5 The dark sector's edge, and two controls

In the target law the phantom of each top-level system ends at an edge $r_e = x_e r_{\text{ta}}$, a fixed fraction of the system's turnaround radius (CFG4). Two data sets pull on x_e .

- **KiDS-1000 lensing** around isolated galaxies [Bro21] prefers $x_e \approx 0.62$. Scored with the programme's machinery (FP20's exact projector, a linear 2-halo term whose amplitude is capped at each lens's peak-background bias), it beats the untruncated law by $\Delta\chi^2 = -35$ to -38 (CFG21).
- **The cosmic cold budget** limits the phantoms summed over the stellar mass function (GAMA, with xGASS gas fractions [Cat18]) to the cold matter that has turned around. Owners are counted at the level the switch specifies: turned-around associations, linked at each system's self-consistent turnaround radius (CFG24). The budget then allows $x_e \leq 0.390/0.336$. KiDS's cost at that edge, against its own best, is $\Delta\chi^2 = 29.4$ and 32.9 for $P2$ and ν_{mono} (5.4 – 5.7σ) canonical, and 45 – 49 alternative.

CFG24 also found an error in the budget's galaxy sample. The Milky Way's own group sits at 0.12 Mpc in the catalogue's velocity-distance convention [KT17], where its integrated magnitude becomes a spurious $4.4 \times 10^{12} M_\odot$. Removing it moves every budget edge up by about 0.006 ; the correction is recorded against the earlier lanes.

Control 1: standard halos in the same machinery. NFW halos with a standard concentration relation [Duf08] and a 2-halo term capped at the halo bias [Tin10] misfit the same KiDS data, $\chi^2 = 174$ against the framework's 104.7 . The misfit sits at $R \geq 0.6$ Mpc, the vanilla halo model's transition region (CFG23; the projection is exact against the analytic NFW profile [WB00] to 3×10^{-4}). A 30% lower concentration with an extended one-halo term reaches $\chi^2 = 99.3$, so KiDS does not prefer either model. The same spherical-infall engine that killed an inflow-derived edge for the framework (FG016) also misfits standard halos, at 149 – 173 (CFG25). The smallest spread the machinery shows for standard physics under reasonable modelling choices is $\Delta\chi^2 = 50.2$, above the framework's canonical budget cost. The budget-versus-lensing tension is therefore *not decidable with this machinery*, and it is not an exclusion (CFG27).

Control 2: the Local Group. The zero-velocity radius the programme adopts for the Local Group is 0.93 ± 0.12 Mpc: the stack of 14 nearby groups of [KK18] (0.93 ± 0.02 ; the Local Group alone 0.91 ± 0.05), with the uncertainty the programme's re-analysis of their data found (lane FP18). With it, the Local Group excludes a universal sharp edge jointly with KiDS at 5 – 7σ (CFG21). But standard halos fitted to the same lenses, placed in the same timing model, fare worse: $T = 24.5$ – 72.5 against the framework's 26 – 28 (CFG23). Lensing masses of isolated Milky Way- and M31-mass galaxies put a standard-halo Local Group at $R_0 = 1.5$ – 1.9 Mpc. The spherical zero-velocity radius does not discriminate between the two models.

6 The prediction for Gaia DR4

A wide binary forms embedded in the Milky Way, so under ownership it owns no phantom and is Newtonian outright. The programme’s frozen Gaia DR4 pre-registration fixes the estimator in advance: 16 cuts, a Newtonian forward model, the statistic $\hat{\gamma}$ and its decision rows. It registered this reading as Arm C (Amendment 13, filed 2026-09-27):

$$\gamma_v = 1.000 \text{ exactly,} \quad (2)$$

on both footings, and independent of a_0 , the kernel, the external field and any coherence length. Arm C is **falsified** at $\hat{\gamma} \geq 1.084$ if the frozen stability requirements pass: every variant of the strictness ladder within $1\sigma_{\text{fit}}$, the ratio κ of the estimator in $[0.95, 1.05]$, and the probe with the non-single-star screen off moving $\hat{\gamma}$ toward the verdict value. No verdict is given above 1.23.

Two other readings of the same law are registered beside it:

- **Arm A**, the kernel as standard MOND with the Galactic external field, predicts 1.1614–1.1814 canonical and 1.1917–1.2267 alternative. A Newtonian result falsifies it. Arms A and C cannot both survive: they sit $5.8/6.8\sigma_{\text{tot}}$ apart.
- **The derivation chain’s filtered law** (Amendment 14, filed 2026-09-28) has a ceiling of 1.0725/1.0900 and is falsified at $\hat{\gamma} \geq 1.157/1.174$.

The test therefore cuts both ways among the framework’s own readings. A MOND-like boost falsifies ownership and the chain’s law. A Newtonian result falsifies standard-MOND wide binaries and leaves ownership standing, but it does not confirm ownership over Newton, which makes the same prediction.

The catalogue chain applies the published method of El-Badry, Rix & Heintz [EBRH21] and the frozen cuts, including those of Banik et al. [Ban24]. It has been run end to end on DR3, recovering 99.4% of their published pairs; the DR3 values of $\hat{\gamma}$ are non-scoring by rule. DR4 reorganises its source catalogue into a curated table and several all-source tables. A draft amendment fixing the mapping of the frozen cuts onto those tables is prepared; if it is not filed before the release, that mapping will be reported as a post-release choice.

7 What is not claimed

- $\kappa = \frac{1}{2}$ is fitted. 32π has no geometric content beyond the factor 4: no natural construction on the vacuum gives $\frac{1}{2}$, and the data ($\kappa = 0.530 \pm 0.037$ combined) do not single it out among simple constants (k05).
- The dark component’s *mass* is required: a cold fluid with the cosmic amount. It is read as a state of the framework’s own field, not a new particle; this note does not supply its microphysics.
- The Milky Way’s ultra-faint satellites are not explained.
- The relativistic completion is not complete. The programme’s surviving covariant action has an obstructed region gate.
- Nothing here shows the framework is preferred over Λ CDM by any data set. Both lensing controls above say the present machinery cannot decide between them.

8 Reproducibility

Every number is produced by a committed script. Each lane was run first with a mutation that must make its headline fail, then in its main mode, with separate outputs. The lanes are in `campaign_fresh_gravity/` and the pre-registration is in `prep_2026/gaia_dr4_prep/`.

lane	script (commit)
target law, SPARC fits	CFG4_target.py, CFG4_galaxy_law.py (3409d12da)
ownership (FG001)	CFG7_hierarchy_fg001.py (3bdf877d)
Chae refit	CFG8_chae_kernel.py (302ff4bd9)
local virial theorem	CFG9_local_virial.py (bbc4200cf)
kernel shape, calibrated	CFG14_shape_calibration.py (8cedf6757)
infall baryons / harness	CFG18_satellite_infall_gas.py (f078e63cf), CFG19_harness_rescore.py (8ace2ee52)
KiDS-Local Group	CFG21_kids_lg_joint.py (d7093c210)
Λ CDM controls	CFG23_lcdm_control.py (27a43a743), CFG25_fg016_lcdm_control.py (1f51ee1e3)
budget, associations / closure	CFG24_budget_associations.py (bf27d136c), CFG27_edge_thread_closure.py (5aa9ad508)
κ geometry	kappa_closure/k05_is_32pi_special.py (937cca01b)
DR4 Arm C / chain ceiling	PREREGISTRATION_DR4.md, Amendments 13 (fc64ffaac) and 14 (ef0556b4e)

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